

Malaria Detection

Deep learning model for red blood cell image classification

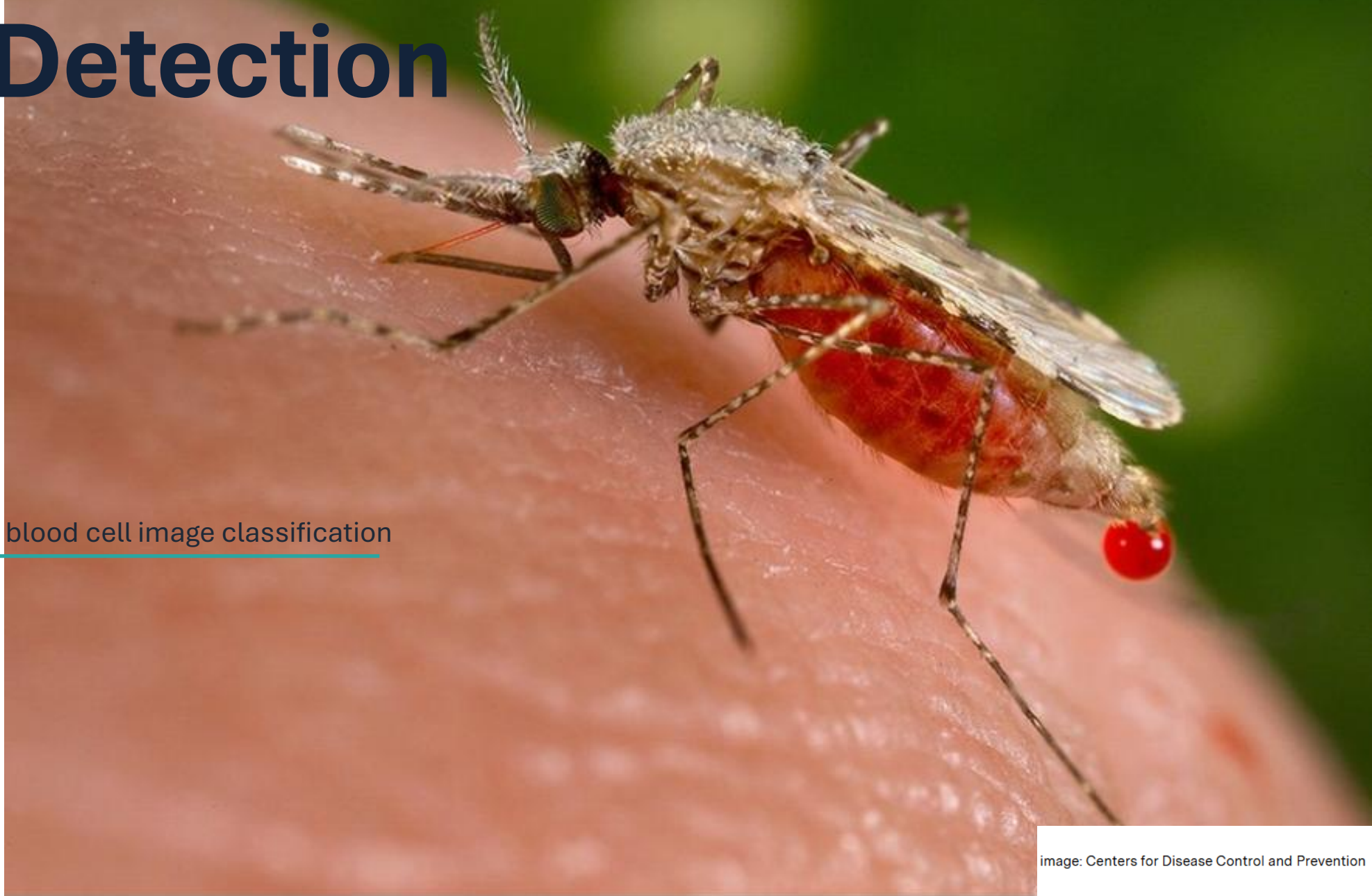


image: Centers for Disease Control and Prevention

Malaria Detection

Deep learning model for red blood cell image classification

Global Health Issue

282M

estimated
cases
in 2024

610K

estimated
deaths
in 2024

WHO 2025

- **Clinical burden:** Plasmodium parasites infect red blood cells and can lead to severe complications or death.
- **Most vulnerable:** Children under five remain among the highest-risk groups for malaria-related mortality.



image: Centers for Disease Control and Prevention

Problem: malaria screening is image-intensive

Manual microscopy is effective, but workflow performance depends on expert time and consistency.

Operational bottleneck

Blood smear review can be time-consuming in high-volume settings and is dependent on trained personnel.

Clinical risk

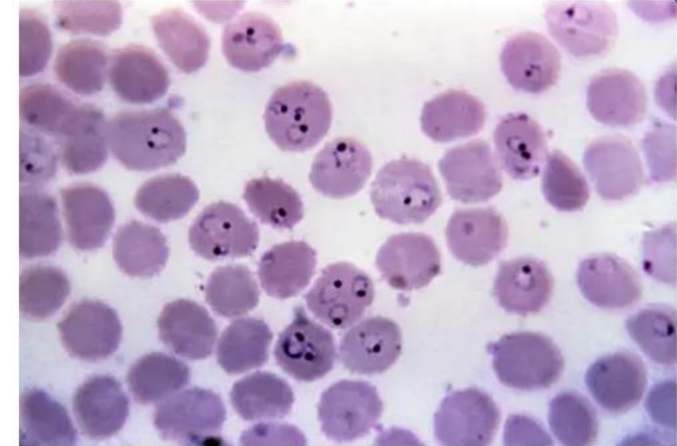
False negatives matter most: a parasitized cell labeled as uninfected represents a missed infected cell.

Business need

A scalable decision-support system can triage cell images, reduce manual workload, and support more consistent review.

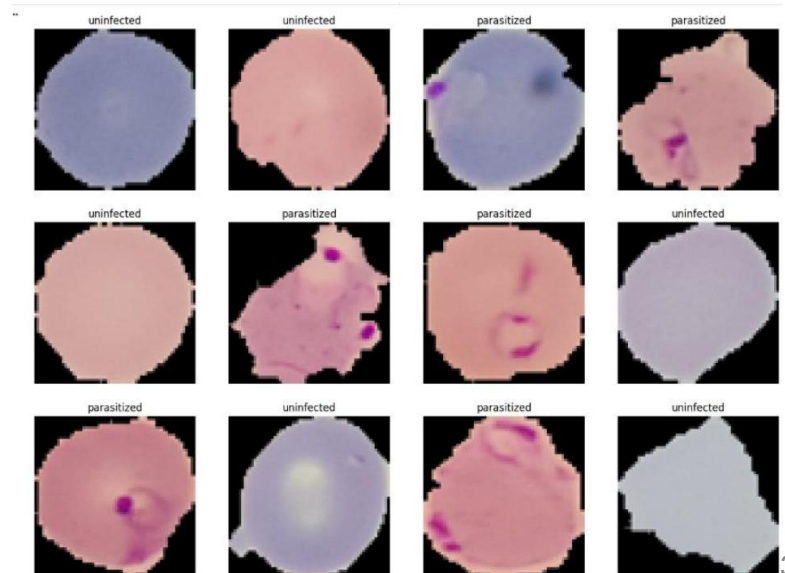
●Need for a scalable and reliable screening-support process that can reduce missed infected cells.

Decision-support, not replacement



Picture of red blood cells infected with malaria parasites. The parasites look like rings inside the cells. SOURCE: CDC/Steven Glenn, Laboratory & Consultation Division

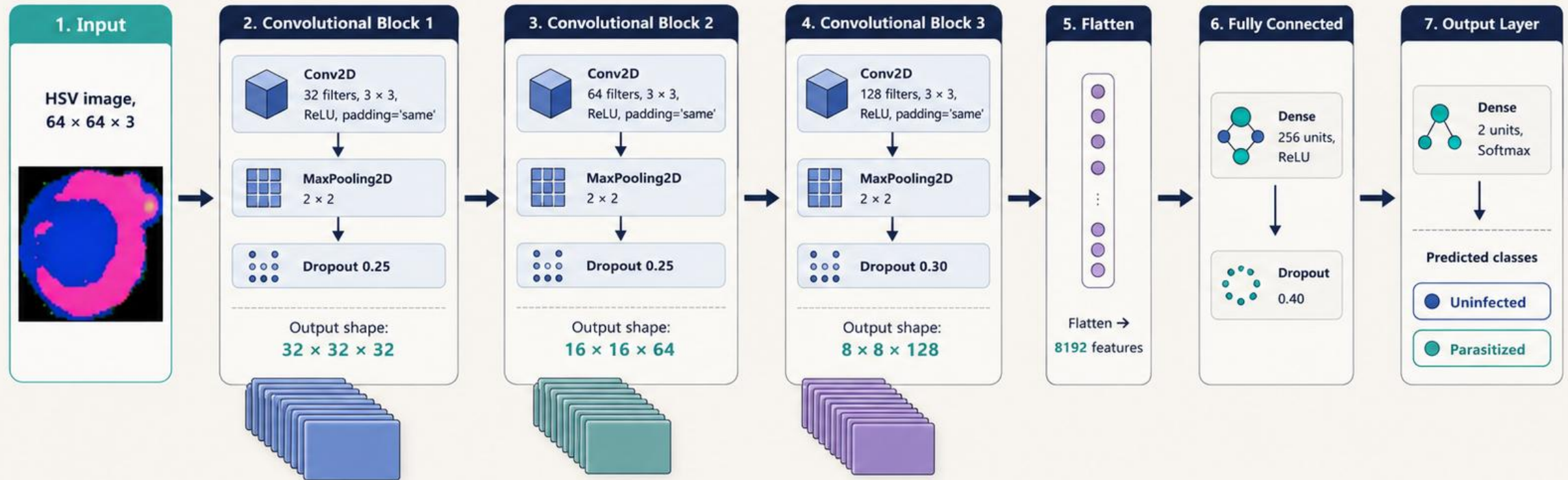
●The goal is to develop a deep learning model that can help with the early and accurate detection of malaria from red blood cell images. Specifically, the model should classify each cell image as either parasitized or uninfected.



HSV CNN Architecture

Final model for malaria red blood cell image classification

Solution



Optimizer:

Adam (learning rate = 0.001)



Loss:

binary cross-entropy



Metric:

accuracy

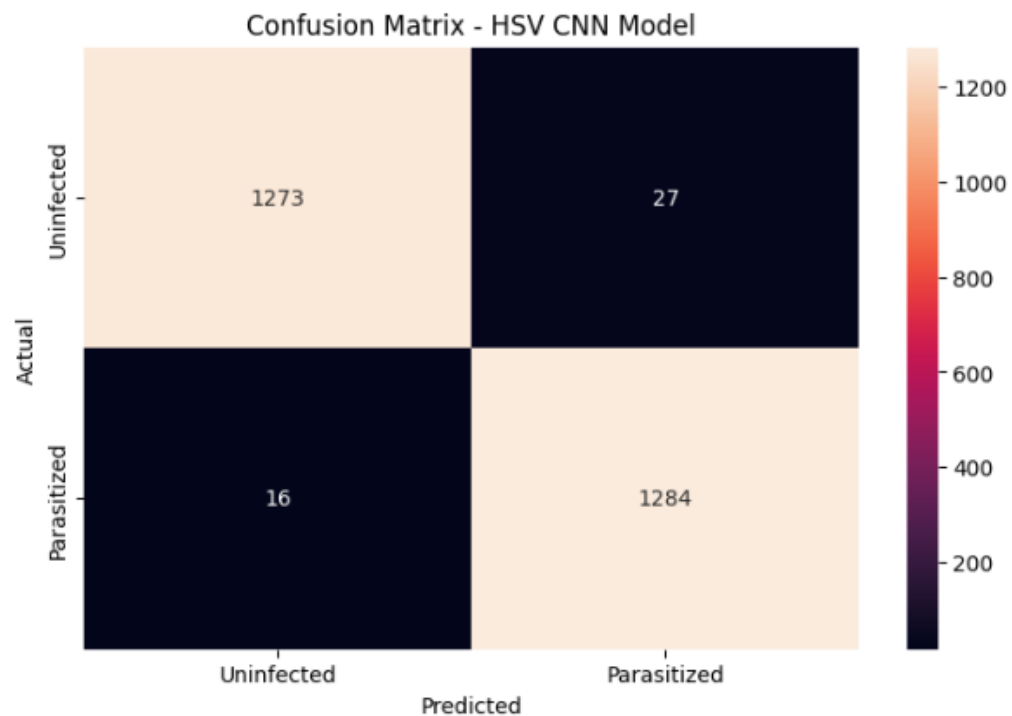


Early stopping and model checkpoint used during training

Final HSV CNN performance

high accuracy with low missed parasitized cells.

	precision	recall	f1-score	support
Uninfected	0.99	0.98	0.98	1300
Parasitized	0.98	0.99	0.98	1300
accuracy			0.98	2600
macro avg	0.98	0.98	0.98	2600
weighted avg	0.98	0.98	0.98	2600



Best balance of accuracy and parasitized-cell detection

98.35%

test accuracy

98.77%

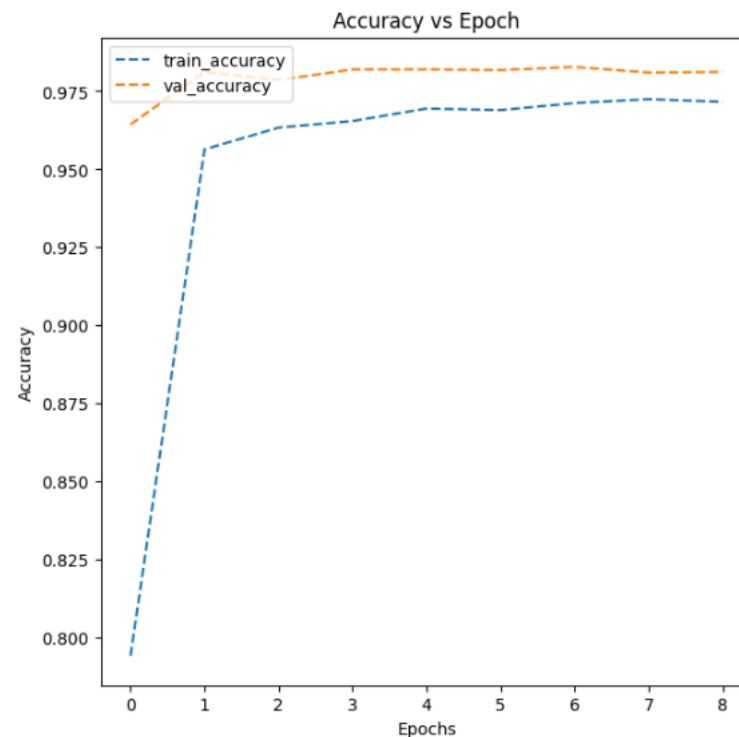
parasitized recall

16

false negatives

27

false positives



Learning curve

Validation accuracy remained stable and slightly above training accuracy, consistent with augmentation/dropout making training harder.



Recommendations for Implementation



1

Deploy the HSV CNN as a decision-support tool for red blood cell image screening, with suspected parasitized cells flagged for expert review.



2

Validate externally using datasets from different staining protocols, microscopes, image resolutions, and laboratory environments before operational use.



3

Benchmark the solution against current state-of-the-art deep learning approaches to confirm whether the HSV CNN remains the most appropriate deployment model.



4

Optimize the Operating Threshold. Prioritize parasitized recall when setting the operating threshold, because false negatives are the most critical error in malaria screening.



5

Implement in Phases and Monitor Performance. Integrate into a controlled workflow where parasitized predictions are reviewed first, uninfected predictions receive quality-control checks, and uncertain cases are escalated. using accuracy, recall, precision, F1-score, confusion matrices, and false-negative counts, with recalibration if performance declines.

Key Actions for Stakeholders

1. Data Science Team

- **Standardize preprocessing:** $64 \times 64 \times 3$ resizing, 0–1 normalization, RGB-to-HSV conversion, and CNN classification.
- **Track performance over time using** accuracy, precision, recall, F1-score, confusion matrices, and false-negative counts.
- **Review misclassified images** to identify artifacts, staining issues, microscope differences, or model limitations.

2. Laboratory and Clinical Stakeholders

- Define how model outputs will be used in practice and how uncertain predictions will be escalated.
- Prioritize parasitized predictions for expert review while keeping uninfected predictions subject to quality-control checks.
- Maintain expert accountability while using the model to improve screening efficiency.

3. Operational Leadership Team

- Evaluate **infrastructure:** image storage, secure data transfer, model hosting, reporting interfaces, access controls, and staff training.
- Define post-deployment ownership for monitoring, retraining, validation, documentation, and approval of model updates.

4. Pilot Implementation

- Run a controlled pilot before scaling the solution broadly.
- **Measure screening time**, flagged-image volume, reviewer workload, false-negative rate, turnaround time, and user acceptance.

Expected Business and Operational Impact



1. Workflow Efficiency

- Supports a scalable and consistent malaria screening workflow.
- Automatically classifies red blood cell images and prioritizes likely infected cells for expert review.
- Reduces time required to screen large image volumes.



2. Screening Performance

- HSV CNN achieved ~98.35% test accuracy and ~98.77% parasitized recall.
- Only 16 false negatives out of 1,300 parasitized test images.
- Low false negatives support the goal of reducing missed infections.



3. Business and Operational Benefits

- Reduces manual workload and supports faster image triage.
- Improves consistency across reviewers through standardized first-pass screening.
- **Valuable in high-volume** laboratories or settings **with limited microscopy expertise.**



4. Resource Allocation and Deployment

- Directs expert attention toward images most likely to require confirmation.
- Supports better use of staff time, faster turnaround, and more consistent screening performance.
- Should be used as decision support, with expert review and external validation before deployment.

Key Risks and Challenges

1. Limited Generalization

- **Images from different laboratories**, microscopes, staining protocols, or patient populations **may differ from the training data**.
- If those differences are substantial, model performance may decline; this risk should be addressed through external validation and ongoing performance monitoring.

2. Overreliance on Model Predictions

- **The model should not be used as a standalone diagnostic system without expert confirmation.**
- False negatives remain clinically important and false positives may increase review workload, so model outputs should remain within a controlled decision-support workflow.

3. Operational and Technical Challenges

- Deployment requires **consistent image quality**, standardized preprocessing, monitoring for model drift, and clear responsibility for reviewing flagged cases.
- **Microscope variability, staining quality, and image artifacts** may also affect performance and should be monitored routinely.

4. Governance and Compliance

- The organization will **need governance procedures for data privacy, model updates, documentation, and clinical validation** before real-world deployment.

Further Analysis Needed

Before real-world deployment, the HSV CNN should be benchmarked, validated, and tested in an operational pilot.

01 Benchmark architectures

Compare HSV CNN with ResNet, EfficientNet, MobileNet, DenseNet, Inception, and microscopy-specific models using recall, false negatives, speed, complexity, and deployment feasibility.

03 Assess robustness across splits

Use **K-fold cross-validation** or repeated validation splits if compute allows, reducing dependence on a single train-validation partition.

05 Analyze model errors

Systematically review false negatives and other errors to identify issues related to image quality, staining artifacts, morphology, or learned representations.

02 Test preprocessing alternatives

Quantify the added value of HSV and evaluate image size, color space, contrast enhancement, stain normalization, and augmentation strategies.

04 Optimize hyperparameters

Tune learning rate, dropout, batch size, filter configuration, dense-layer size, augmentation parameters, and training duration with focus on parasitized recall.

06 Evaluate confidence and interpretability

Use confidence calibration and visual explanation methods to verify that predictions rely on biologically plausible parasite-related regions.

Analytical priority: confirm the model is reproducible, technically justified, robust, and explainable before broader operational evaluation.

Executive Summary

Final low-code submission · malaria red blood cell image classification

- **Malaria diagnosis relies on microscopy:** effective, but time-consuming, labor-intensive, and reviewer-dependent.
- **Goal:** classify red blood cell images as parasitized or uninfected to support automated screening.
- **Data finding:** balanced classes; infection cues are subtle, localized, and tied to staining, texture, and intracellular patterns.
- **Modeling approach:** compared baseline CNN, modified CNNs, augmentation, VGG16 transfer learning, and HSV CNN.
- **Selected model:** HSV CNN, chosen for the best balance of accuracy and parasitized-cell detection.
- **Implementation:** decision-support workflow with expert review and external validation before deployment.

98.35%

test accuracy

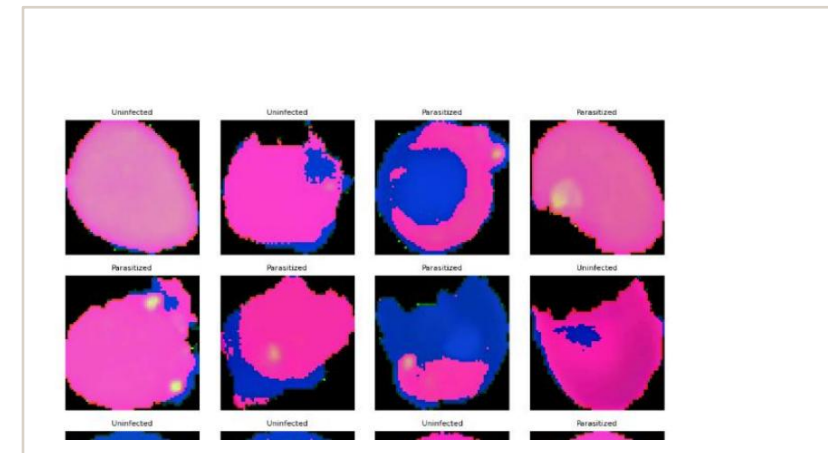
98.77%

parasitized recall

16

false negatives

Final model at a glance



HSV-transformed red blood cell images highlight color and staining patterns

Recommended use

Deploy as a screening decision-support tool: flag likely parasitized cells for expert review, monitor false negatives, and validate across sites before clinical use.